

## SP250 Homework 1

1. Let  $a$  and  $b \neq 0$  be integers, such that  $a$  is divisible by  $b$ . Prove rigorously that for any integer  $c$ ,  $a \cdot c$  is divisible by  $b$  (In other words,  $b \mid a \cdot c$ ).

Solution:

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2. What is the sum of all the odd positive integers less than 1000?

Solution:

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3. The country of Mathlandia has 10 major cities, any two of which are connected by exactly one road. How many roads are there in total between those 10 cities?

Solution:

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4. The numbers 1, 2, ..., 11, 12 are written in order along the edge of a round disk (like a watch). Cut the disk into 3 pieces (by straight lines) so that the sum of numbers in each piece is the same.

Solution:

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5. All integers between 1 and 999 were written out into one very large number. How many 0's were used? Explain your reasoning.

Solution:

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6. Two piles of stones are given. One pile has 10 stones while the other pile has 33 stones. Two players take turns playing the game where, in turn, each player may take 1,2,3 or 4 stones from one of the piles. The player who takes the last stone wins. Which player has a winning strategy, first or second?

Solution:

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